

Module 5

Valuing Bonds and Stocks

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

From Module 4 to Module 5

- **Module 4**, the time value of money, the present value of future cash flows

From Module 4 to Module 5

- **Module 4**, the time value of money, the present value of future cash flows
- **Module 5**, the same time-value-of-money principles value bonds and stocks
 - Bond, the present value of its coupons and face value
 - Stock, the present value of its expected dividends

Every Financial Asset Is Debt or Equity

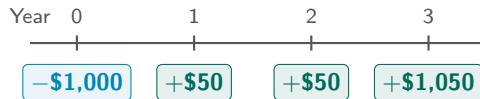
- **Financial asset**, value from a contractual right or ownership claim



Debt's Cash Flow Is Fixed, a Stock's Is Not

Debt

\$1,000, 3 years, 5% interest

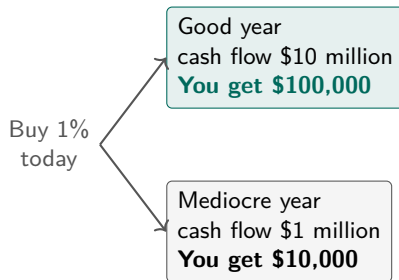


Every payment known in advance

Stock

1% of the company

Next year



Value of claim depends on how the company does

Common Types of Corporate Financing

Debt, lending to the company

Main forms, bonds and loans

Holder gets a fixed promised amount

“Senior”, paid back before equity in
bankruptcy

No voting rights or control

Common Types of Corporate Financing

Debt, lending to the company

Main forms, bonds and loans

Holder gets a fixed promised amount

“Senior”, paid back before equity in bankruptcy

No voting rights or control

Equity, stock ownership in the company

Holder gets dividends

“Residual claimant”, paid whatever is left after all promised payments, no guaranteed payout

Voting rights

Module 5, Part 1

Bonds

Bonds, Definitions

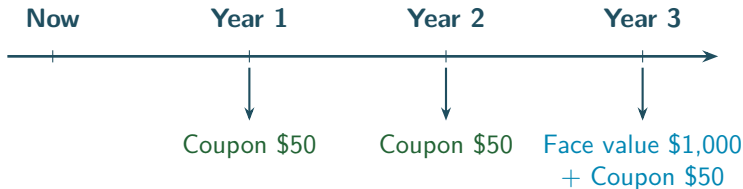
- **Bond**, a security that obligates the issuer (the borrower) to make specified payments to the investor (the lender, or “bondholder”)

Bonds, Definitions

- **Bond**, a security that obligates the issuer (the borrower) to make specified payments to the investor (the lender, or “bondholder”)
- Typically periodic payments (the “coupon”), then a “face value” at the end

Bonds, Definitions

- **Bond**, a security that obligates the issuer (the borrower) to make specified payments to the investor (the lender, or “bondholder”)
- Typically periodic payments (the “coupon”), then a “face value” at the end
- E.g., a three-year bond with an annual coupon of \$50 and a face value of \$1,000, the borrower promises to pay,



Bonds, Terminology

- **Face value**, the final lump-sum payment at the “maturity” of the bond, usually \$1,000 per bond in the U.S.



Imaged by Heritage Auctions, HA.com

Bonds, Terminology

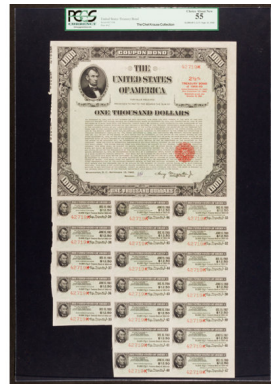
- **Face value**, the final lump-sum payment at the “maturity” of the bond, usually \$1,000 per bond in the U.S.
- **Maturity**, the time when the debt expires (“matures”) and the face value is paid
- **Time to maturity**, the number of years until maturity



Imaged by Heritage Auctions, HA.com

Bonds, Terminology

- **Face value**, the final lump-sum payment at the “maturity” of the bond, usually \$1,000 per bond in the U.S.
- **Maturity**, the time when the debt expires (“matures”) and the face value is paid
- **Time to maturity**, the number of years until maturity
- **Coupon**, the periodic payments to the bondholder before maturity

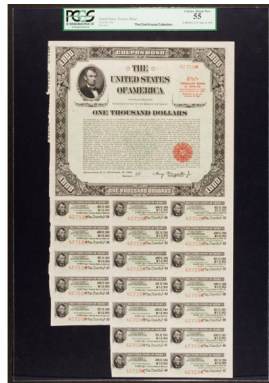


Imaged by Heritage Auctions, HA.com

Bonds, Terminology

- **Face value**, the final lump-sum payment at the “maturity” of the bond, usually \$1,000 per bond in the U.S.
- **Maturity**, the time when the debt expires (“matures”) and the face value is paid
- **Time to maturity**, the number of years until maturity
- **Coupon**, the periodic payments to the bondholder before maturity
- **Coupon rate**,

$$\text{Coupon rate} = \frac{\text{Annual coupon}}{\text{Face value}}$$



Imaged by Heritage Auctions, HA.com

Bonds, Terminology

- Common fallacy, the coupon is not “interest”, and the coupon rate is not necessarily the interest rate

Bonds, Terminology

- Common fallacy, the coupon is not “interest”, and the coupon rate is not necessarily the interest rate
 - E.g., **Module 4**'s Puerto Rico bond, the coupon was 0 but the interest rate was 5.15%

Example, Puerto Rico Borrows

Recall from **Module 4**

- In 2007, Puerto Rico sold bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each
- What was the interest rate paid by Puerto Rico?



Example, Puerto Rico Borrows

Recall from **Module 4**



$$N = 47, \quad I = ?$$

Example, Puerto Rico Borrows

Recall from **Module 4**

- In 2007, Puerto Rico sold bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each
- What was the interest rate paid by Puerto Rico?



- Solution,
 - 47 **N**
 - -94.40 **PV**
 - 0 **PMT**
 - 1000 **FV**
 - **I/Y**

Example, Puerto Rico Borrows

Recall from **Module 4**

- In 2007, Puerto Rico sold bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each
- What was the interest rate paid by Puerto Rico?



- Solution,
 - 47 **N**
 - -94.40 **PV**
 - 0 **PMT**
 - 1000 **FV**
 - **I/Y**
- The rate is 5.15%

Bonds, Pricing

- **Bond value**, the present value of all its cash flows (coupons and face value), discounted at the “required rate of return” r
 - “Required rate of return”, the name for the discount rate in this application

Bonds, Pricing

- **Bond value**, the present value of all its cash flows (coupons and face value), discounted at the “required rate of return” r
 - “Required rate of return”, the name for the discount rate in this application
- E.g., for a bond with time to maturity T ,

$$\text{Value} = \text{PV} = \frac{\text{Coupon}_1}{(1+r)^1} + \frac{\text{Coupon}_2}{(1+r)^2} + \dots + \frac{\text{Coupon}_T + \text{Face value}}{(1+r)^T}$$

Bonds, Pricing

- **Bond value**, the present value of all its cash flows (coupons and face value), discounted at the “required rate of return” r
 - “Required rate of return”, the name for the discount rate in this application

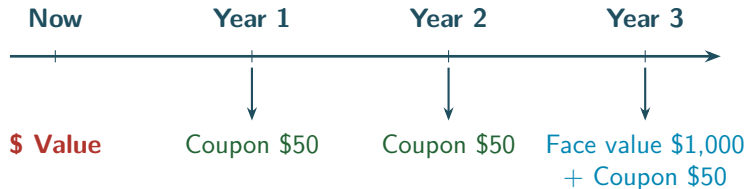
- E.g., for a bond with time to maturity T ,

$$\text{Value} = \text{PV} = \frac{\text{Coupon}_1}{(1+r)^1} + \frac{\text{Coupon}_2}{(1+r)^2} + \dots + \frac{\text{Coupon}_T + \text{Face value}}{(1+r)^T}$$

- Investors choose the opportunity cost, “what could I get investing in something else with a similar risk”, as the required rate of return r

Example, Bond Pricing

- A bond has an annual coupon of \$50 and a face value of \$1,000, with three years to maturity



Example, Bond Pricing

- A bond has an annual coupon of \$50 and a face value of \$1,000, with three years to maturity



- What is the value of the bond if the required rate of return r is 4%?

Example, Bond Pricing

- A bond has an annual coupon of \$50 and a face value of \$1,000, with three years to maturity



- What is the value of the bond if the required rate of return r is 4%?

$$\text{Value} = \frac{\$50}{(1 + 0.04)^1} + \frac{\$50}{(1 + 0.04)^2} + \frac{\$50 + \$1,000}{(1 + 0.04)^3} = \$1,027.75$$

Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$50, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to value the bond

Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$50, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to value the bond
- Solution,
 - 1000 **FV**
 - 50 **PMT**
 - 3 **N**
 - 4 **I/Y**
 - **CPT**, then **PV**

Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$50, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to value the bond
- Solution,
 - 1000 **FV**
 - 50 **PMT**
 - 3 **N**
 - 4 **I/Y**
 - **CPT**, then **PV**
- The output is $-\$1,027.75$ (why the minus sign? recall how financial calculators get to the answer in [Module 4](#))

Valuing Bonds with a Financial Calculator

- The financial calculator keys map onto a bond,
 - **N**, the number of periods (years) until maturity
 - **PV**, the value of the bond (the present value of the coupons and the face value)
 - **PMT**, the coupons (in dollars)
 - **FV**, the face value (the lump sum paid at the end)
 - **I/Y**, the discount rate (the required rate of return r)

Recap of Part 1

- Companies issue bonds (and other debt) to raise financing from investors

Recap of Part 1

- Companies issue bonds (and other debt) to raise financing from investors
- In return, the companies make specified payments of coupons and face value to the investors

Recap of Part 1

- Companies issue bonds (and other debt) to raise financing from investors
- In return, the companies make specified payments of coupons and face value to the investors
- The value of a bond is the discounted payments, at the required rate of return

Module 5, Part 2

Bond Yields

Bond Yields

- **Yield**, the discount rate y for which the present value of the bond's payments (coupons and face value) equals its price

Bond Yields

- **Yield**, the discount rate y for which the present value of the bond's payments (coupons and face value) equals its price
- Given the price of a bond (PV), its maturity T , and its coupons, find the yield by solving for y ,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \cdots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

Example, Puerto Rico Borrows

Recall from **Module 4**

- In 2007, Puerto Rico sold bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each
- What was the interest rate paid by Puerto Rico?



- Solution,
 - 47 **N**
 - -94.40 **PV**
 - 0 **PMT**
 - 1000 **FV**
 - **I/Y**
- The rate is 5.15%

Example, Puerto Rico Borrows

Recall from **Module 4**

- In 2007, Puerto Rico sold bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each
- What was the interest rate paid by Puerto Rico?
- **Here we are calculating the yield!**



- Solution,
 - 47 **N**
 - -94.40 **PV**
 - 0 **PMT**
 - 1000 **FV**
 - **I/Y**
- The rate is 5.15%

Example, Bond Yields

- Solving for the yield analytically is hard, so use the financial calculator
- A bond's annual coupon rate is 5%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,027.75. What is its yield?

Example, Bond Yields

- Solving for the yield analytically is hard, so use the financial calculator
- A bond's annual coupon rate is 5%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,027.75. What is its yield?
- Solution,
 - 1000 **FV**
 - 50 **PMT**
 - 3 **N**
 - -1027.75 **PV**
 - **CPT**, then **I/Y**

Example, Bond Yields

- Solving for the yield analytically is hard, so use the financial calculator
- A bond's annual coupon rate is 5%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,027.75. What is its yield?
- Solution,
 - 1000 **FV**
 - 50 **PMT**
 - 3 **N**
 - -1027.75 **PV**
 - **CPT**, then **I/Y**
- The yield is 4%

Required Rate of Return vs. Yield

Required rate of return r

- The discount rate for computing the bond's value
- Chosen by investors as the opportunity cost, the return on alternatives with a similar risk
- Input r , output the value
(**CPT** **PV**)

Yield y

Required Rate of Return vs. Yield

Required rate of return r

- The discount rate for computing the bond's value
- Chosen by investors as the opportunity cost, the return on alternatives with a similar risk
- Input r , output the value
(**CPT** **PV**)

Yield y

- The discount rate for which the present value of the payments equals the price
- Implied by the market price
- Input the price, output the yield
(**CPT** **I/Y**)

Required Rate of Return vs. Yield

Required rate of return r

- The discount rate for computing the bond's value
- Chosen by investors as the opportunity cost, the return on alternatives with a similar risk
- Input r , output the value
(**CPT** **PV**)
- **Subjective**

Yield y

- The discount rate for which the present value of the payments equals the price
- Implied by the market price
- Input the price, output the yield
(**CPT** **I/Y**)
- **Objective**

Practice Problem

- The table shows data for three bonds. Each bond's coupon is zero. Each bond's face value is \$1,000

Bond	Price	Maturity (years)	Yield
A	\$310	25	?
B	\$310	?	9%
C	?	11	8%

- What is the yield of bond A?
- What is the maturity of bond B?
- What is the price of bond C?

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	?	-310	0	1,000
b. Bond B	?	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	?	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

- a. The yield of bond A is 4.796%

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	13.59	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

- a. The yield of bond A is 4.796%
- b. The maturity of bond B is 13.59 years

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

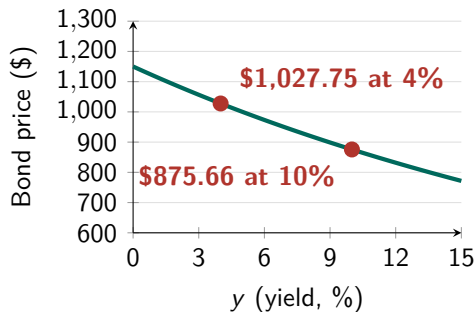
	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	13.59	9	-310	0	1,000
c. Bond C	11	8	428.88	0	-1,000

- a. The yield of bond A is 4.796%
- b. The maturity of bond B is 13.59 years
- c. The price of bond C is \$428.88

Bond Prices and Yields Move in Opposite Directions

- Because the bond yield is the y such that,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \dots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

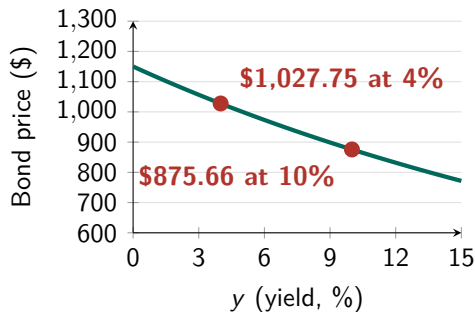


Bond Prices and Yields Move in Opposite Directions

- Because the bond yield is the y such that,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \dots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

- Higher y , lower bond price (the chart shows the running example bond)

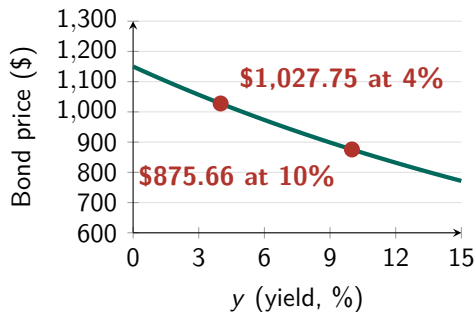


Bond Prices and Yields Move in Opposite Directions

- Because the bond yield is the y such that,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \dots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

- Higher y , lower bond price (the chart shows the running example bond)
- Bond prices and yields move in opposite directions!



What Determines Bond Price (and Therefore Yield)?

1. Market (risk-free) interest rates (time value of money)

What Determines Bond Price (and Therefore Yield)?

1. Market (risk-free) interest rates (time value of money)
2. Risk of default (compensation for taking risks)
 - The risk that a bond issuer may default on its bonds
 - **Default premium**, the additional yield on a bond that investors require for bearing default risk

What Determines Bond Price (and Therefore Yield)?

1. Market (risk-free) interest rates (time value of money)
2. Risk of default (compensation for taking risks)
 - The risk that a bond issuer may default on its bonds
 - **Default premium**, the additional yield on a bond that investors require for bearing default risk

$$\text{Bond yield} = \text{risk-free rate} + \text{default premium}$$

Two Borrowers, Two Yields

- Borrower, the federal government
- Default risk, none
- Bond yield = risk-free rate
- Yield is low and price is high



Two Borrowers, Two Yields

- Borrower, the federal government
 - Default risk, none
 - Bond yield = risk-free rate
 - Yield is low and price is high
-
- Borrower, a small corporation with risky cash flows
 - Default risk, high
 - Bond yield = risk-free rate + default premium
 - Yield is high and price is low



Default Risk, Credit Ratings

- Credit rating agencies provide ratings intended to capture default risk
 - Largest, Moody's, Standard & Poor's, and Fitch

Default Risk, Credit Ratings

- Credit rating agencies provide ratings intended to capture default risk
 - Largest, Moody's, Standard & Poor's, and Fitch
- Bond ratings depend on,
 1. The probability of default
 2. The expected loss in default

Default Risk, Credit Ratings

- Credit rating agencies provide ratings intended to capture default risk
 - Largest, Moody's, Standard & Poor's, and Fitch
- Bond ratings depend on,
 1. The probability of default
 2. The expected loss in default
- **Investment-grade bonds**, rated Baa or higher by Moody's, or BBB or higher by Standard & Poor's

Default Risk, Credit Ratings

- Credit rating agencies provide ratings intended to capture default risk
 - Largest, Moody's, Standard & Poor's, and Fitch
- Bond ratings depend on,
 1. The probability of default
 2. The expected loss in default
- **Investment-grade bonds**, rated Baa or higher by Moody's, or BBB or higher by Standard & Poor's
- **High-yield bonds** ("junk" bonds), rated Ba or BB and lower

Default Risk, Credit Ratings

Moody's	S&P	Default within 10 yr
Investment-grade bonds		
Aaa	AAA	0.1%
Aa	AA	0.7%
A	A	2.2%
Baa	BBB	3.5%
High-yield bonds		
Ba	BB	15.7%
B	B	35.5%
Caa, Ca	CCC, CC	48.9%
C	C	n/a

Source, Brealey, Myers, and Marcus, Table 6.2

Default Risk, Credit Ratings

Moody's	S&P	Default within 10 yr
Investment-grade bonds		
Aaa	AAA	0.1%
Aa	AA	0.7%
A	A	2.2%
Baa	BBB	3.5%
High-yield bonds		
Ba	BB	15.7%
B	B	35.5%
Caa, Ca	CCC, CC	48.9%
C	C	n/a

Source, Brealey, Myers, and Marcus, Table 6.2

- Yields tend to be higher (prices lower) for lower-rated bonds

Treasury Yields at 20-Year Highs

- **Treasury yield**, the yield on bonds issued by the U.S. government (“Treasury bonds”)



PBS NewsHour, August 20, 2026 (7:14), “Why Treasury yields are at 20-year highs”

Recap of Part 2

- Yield, the discount rate for which the present value of the bond's payments equals its price

Recap of Part 2

- Yield, the discount rate for which the present value of the bond's payments equals its price
- Bond prices and yields move in opposite directions

Recap of Part 2

- Yield, the discount rate for which the present value of the bond's payments equals its price
- Bond prices and yields move in opposite directions
- Prices fall and yields rise with market interest rates and with default risk, lower-rated bonds carry higher yields

Module 5, Part 3

Stock Valuation, Dividend Discount Model

Valuing Stocks

- Valuing stocks helps us decide what to invest in

Valuing Stocks

- Valuing stocks helps us decide what to invest in
- Two ways to value stocks,
 1. Dividend discount model
 2. Valuation by comparables

Valuing Stocks

- Valuing stocks helps us decide what to invest in
- Two ways to value stocks,
 1. Dividend discount model
 2. Valuation by comparables
- The approaches are complementary, no approach provides “the” right answer
- Professionals use a combination

Stock Valuation, Terminology

- **Dividend**, periodic cash paid out to shareholders

Stock Valuation, Terminology

- **Dividend**, periodic cash paid out to shareholders
- **Earnings per share (EPS)**, $\frac{\text{Net income}}{\text{Number of shares outstanding}}$

Stock Valuation, Terminology

- **Dividend**, periodic cash paid out to shareholders
- **Earnings per share (EPS)**, $\frac{\text{Net income}}{\text{Number of shares outstanding}}$
- **P/E ratio**, $\frac{\text{Stock price}}{\text{Earnings per share}}$

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P)?

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P)?
- Stock value, the present value of the cash flows you get from owning it,

$$P = \text{PV}(\text{Dividends}) = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty}$$

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P)?

- Stock value, the present value of the cash flows you get from owning it,

$$P = \text{PV}(\text{Dividends}) = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty}$$

- **Required return on equity** (r_e), the discount rate, set by the opportunity cost of investing in something with similar risk

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P)?

- Stock value, the present value of the cash flows you get from owning it,

$$P = \text{PV}(\text{Dividends}) = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \dots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty}$$

- **Required return on equity** (r_e), the discount rate, set by the opportunity cost of investing in something with similar risk
- $\text{Div}_1, \text{Div}_2, \dots$ are *expected* dividends

Gordon Growth Model

- Special cases, using **Module 4**'s perpetuity formulas

Gordon Growth Model

- Special cases, using **Module 4**'s perpetuity formulas
- If dividends are expected to be constant,

$$P = \frac{\text{Div}_1}{r_e}$$

Gordon Growth Model

- Special cases, using **Module 4**'s perpetuity formulas

- If dividends are expected to be constant,

$$P = \frac{\text{Div}_1}{r_e}$$

- If dividends are expected to grow at a constant rate g ,

$$P = \frac{\text{Div}_1}{r_e - g}$$

Gordon Growth Model

- Special cases, using **Module 4**'s perpetuity formulas

- If dividends are expected to be constant,

$$P = \frac{\text{Div}_1}{r_e}$$

- If dividends are expected to grow at a constant rate g ,

$$P = \frac{\text{Div}_1}{r_e - g}$$

- The second formula is often called the “**Gordon growth model**”

Example, Gordon Growth Model

- 1. A firm will pay a \$3 dividend forever (the first dividend exactly a year from now). Its required return on equity is 12%. What will investors pay for the stock?
- 2. What if the firm's stock pays a \$3 dividend a year from now, and then the dividend grows at 8% forever?

Example, Gordon Growth Model

1. Constant dividend



Example, Gordon Growth Model

1. Constant dividend



2. Dividend growing at 8% a year



Example, Gordon Growth Model

- 1. A firm will pay a \$3 dividend forever (the first dividend exactly a year from now). Its required return on equity is 12%. What will investors pay for the stock?
- 2. What if the firm's stock pays a \$3 dividend a year from now, and then the dividend grows at 8% forever?
- Solution, apply the perpetuity formulas,

$$1. \quad P = \frac{\text{Div}_1}{r_e} = \frac{\$3}{0.12} = \$25$$

Example, Gordon Growth Model

- 1. A firm will pay a \$3 dividend forever (the first dividend exactly a year from now). Its required return on equity is 12%. What will investors pay for the stock?
- 2. What if the firm's stock pays a \$3 dividend a year from now, and then the dividend grows at 8% forever?
- Solution, apply the perpetuity formulas,

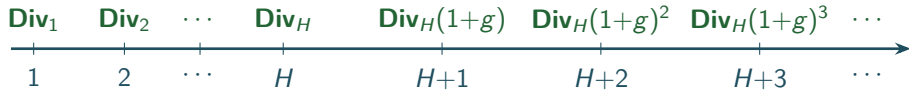
$$1. \quad P = \frac{\text{Div}_1}{r_e} = \frac{\$3}{0.12} = \$25$$

$$2. \quad P = \frac{\text{Div}_1}{r_e - g} = \frac{\$3}{0.12 - 0.08} = \$75$$

Valuing Dividends with Non-Constant Growth

- A common practice is to estimate the dividends year by year over a horizon of H years, then assume constant growth after year H ,

Valuing Dividends with Non-Constant Growth



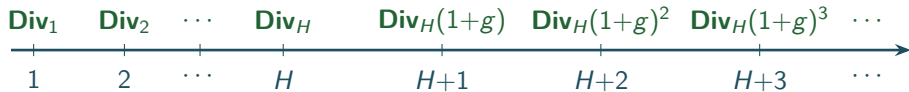
Valuing Dividends with Non-Constant Growth

- A common practice is to estimate the dividends year by year over a horizon of H years, then assume constant growth after year H ,

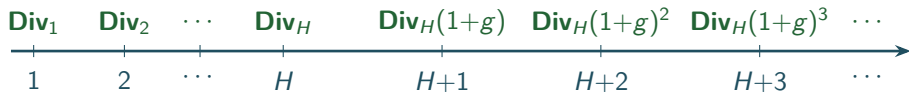
$$PV = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \cdots + \frac{\text{Div}_H}{(1 + r_e)^H} + \frac{P_H}{(1 + r_e)^H}$$

$$P_H = \frac{\text{Div}_H \times (1 + g)}{r_e - g}$$

Valuing Dividends with Non-Constant Growth



Valuing Dividends with Non-Constant Growth



+ P_H

$$\frac{\text{Div}_H \times (1 + g)}{r_e - g}$$

Valuing Dividends with Non-Constant Growth

- A common practice is to estimate the dividends year by year over a horizon of H years, then assume constant growth after year H ,

$$PV = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \cdots + \frac{\text{Div}_H}{(1 + r_e)^H} + \frac{P_H}{(1 + r_e)^H}$$

$$P_H = \frac{\text{Div}_H \times (1 + g)}{r_e - g}$$

- Why? In practice, near-term dividends can be estimated carefully one year at a time, the distant future cannot

Example, Valuing Non-Constant Growth

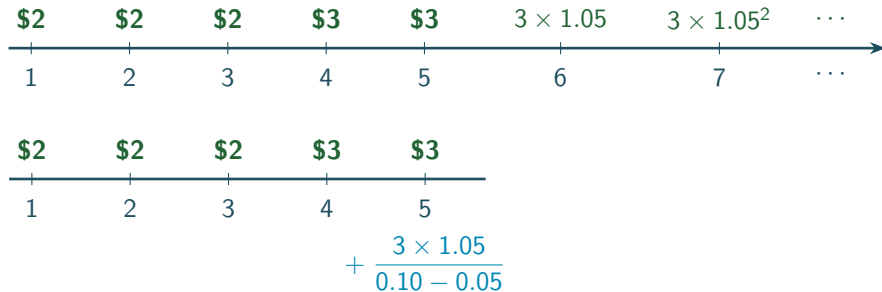
- A stock's expected dividends are in the table. Its required return on equity is 10%. Its dividends grow at 5% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	\$2	\$2	\$2	\$3	\$3

Example, Valuing Non-Constant Growth



Example, Valuing Non-Constant Growth



Example, Valuing Non-Constant Growth

- A stock's expected dividends are in the table. Its required return on equity is 10%. Its dividends grow at 5% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	\$2	\$2	\$2	\$3	\$3

- Solution, discount the five dividends, then the year-5 price of the constant-growth stream

Example, Valuing Non-Constant Growth

- A stock's expected dividends are in the table. Its required return on equity is 10%. Its dividends grow at 5% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	\$2	\$2	\$2	\$3	\$3

- Solution, discount the five dividends, then the year-5 price of the constant-growth stream

$$PV_{\text{Dividends 1-5}} = \frac{2}{1.10} + \frac{2}{1.10^2} + \frac{2}{1.10^3} + \frac{3}{1.10^4} + \frac{3}{1.10^5} = \$8.89$$

Example, Valuing Non-Constant Growth

- A stock's expected dividends are in the table. Its required return on equity is 10%. Its dividends grow at 5% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	\$2	\$2	\$2	\$3	\$3

- Solution, discount the five dividends, then the year-5 price of the constant-growth stream

$$PV_{\text{Dividends 1-5}} = \frac{2}{1.10} + \frac{2}{1.10^2} + \frac{2}{1.10^3} + \frac{3}{1.10^4} + \frac{3}{1.10^5} = \$8.89$$

$$PV_{\text{Price in year 5}} = \frac{3 \times 1.05}{0.10 - 0.05} \times \frac{1}{1.10^5} = \frac{63}{1.10^5} = \$39.12$$

Example, Valuing Non-Constant Growth

- A stock's expected dividends are in the table. Its required return on equity is 10%. Its dividends grow at 5% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	\$2	\$2	\$2	\$3	\$3

- Solution, discount the five dividends, then the year-5 price of the constant-growth stream

$$PV_{\text{Dividends 1-5}} = \frac{2}{1.10} + \frac{2}{1.10^2} + \frac{2}{1.10^3} + \frac{3}{1.10^4} + \frac{3}{1.10^5} = \$8.89$$

$$PV_{\text{Price in year 5}} = \frac{3 \times 1.05}{0.10 - 0.05} \times \frac{1}{1.10^5} = \frac{63}{1.10^5} = \$39.12$$

$$PV_{\text{Stock}} = 8.89 + 39.12 = \$48.00$$

Practice Problem

- Consider the following two stocks,
 - a. Stock A is expected to pay a dividend of \$10 a share forever
 - b. Stock B is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 4% a year forever
- Each stock's required return on equity is 10%. Which stock is the most valuable?

Practice Problem

Stock A



Practice Problem

Stock A



Stock B



Practice Problem

- Solution, both stocks are perpetuities, apply the perpetuity formulas,

$$P_A = \frac{\text{Div}_1}{r_e} = \frac{\$10}{0.10} = \$100$$

Practice Problem

- Solution, both stocks are perpetuities, apply the perpetuity formulas,

$$P_A = \frac{\text{Div}_1}{r_e} = \frac{\$10}{0.10} = \$100$$

$$P_B = \frac{\text{Div}_1}{r_e - g} = \frac{\$5}{0.10 - 0.04} = \$83.33$$

Practice Problem

- Solution, both stocks are perpetuities, apply the perpetuity formulas,

$$P_A = \frac{\text{Div}_1}{r_e} = \frac{\$10}{0.10} = \$100$$

$$P_B = \frac{\text{Div}_1}{r_e - g} = \frac{\$5}{0.10 - 0.04} = \$83.33$$

- Stock A is the most valuable

Recap of Part 3

- Dividend discount model, the stock value is the present value of its expected dividends at the required return on equity r_e

Recap of Part 3

- Dividend discount model, the stock value is the present value of its expected dividends at the required return on equity r_e
- Constant dividends, $P = \text{Div}_1 / r_e$, constant growth (Gordon growth model), $P = \text{Div}_1 / (r_e - g)$

Module 5, Part 4

Stock Valuation, Comparables

Valuing Stocks, Comparables

- What is one of the most common approaches to valuing real estate?

Valuing Stocks, Comparables

- What is one of the most common approaches to valuing real estate?
- Comparables (“comps”)!

Valuing Stocks, Comparables

- What is one of the most common approaches to valuing real estate?
- Comparables (“comps”)!
- E.g., a 2,000 sqft house in New Orleans sells for \$400,000. What is your estimate for a neighboring house of 2,500 sqft?

Valuing Stocks, Comparables

- Real estate comparables use price per square foot. For companies, the most common metric is P/E (price to EPS)

Valuing Stocks, Comparables

- Real estate comparables use price per square foot. For companies, the most common metric is P/E (price to EPS)
- E.g., Company A has twice the earnings per share of Company B, and Company B trades at \$10. What price should Company A's shares trade at?

Valuing Stocks, Comparables

- Real estate comparables use price per square foot. For companies, the most common metric is P/E (price to EPS)
- E.g., Company A has twice the earnings per share of Company B, and Company B trades at \$10. What price should Company A's shares trade at?
- Underlying idea, we can “scale” prices!

Valuing Stocks by Comparables, Four Steps

- Suppose we want to value Company X

Valuing Stocks by Comparables, Four Steps

- Suppose we want to value Company **X**
- 1. Start with a sample of “comparable” companies (**A**, **B**, **C**, ...) whose business characteristics are similar to Company **X**

Valuing Stocks by Comparables, Four Steps

- Suppose we want to value Company **X**
- 1. Start with a sample of “comparable” companies (**A**, **B**, **C**, ...) whose business characteristics are similar to Company **X**
- 2. Pick a ratio (“multiple”) with the stock price or equity value in the numerator, e.g., P/E or market-to-book

Valuing Stocks by Comparables, Four Steps

- Suppose we want to value Company **X**
 1. Start with a sample of “comparable” companies (**A**, **B**, **C**, ...) whose business characteristics are similar to Company **X**
 2. Pick a ratio (“multiple”) with the stock price or equity value in the numerator, e.g., P/E or market-to-book
 3. Calculate the average of that multiple for the comparables **A**, **B**, **C**, ...

Valuing Stocks by Comparables, Four Steps

- Suppose we want to value Company **X**
1. Start with a sample of “comparable” companies (**A**, **B**, **C**, ...) whose business characteristics are similar to Company **X**
 2. Pick a ratio (“multiple”) with the stock price or equity value in the numerator, e.g., P/E or market-to-book
 3. Calculate the average of that multiple for the comparables **A**, **B**, **C**, ...
 4. Back out Company **X**’s implied value as the average multiple times Company **X**’s denominator

$$P_X = (\text{Average P/E of comparables}) \times \text{EPS}_X$$

Example, Valuation by Comparables

- Furniture manufacturer Herman Miller has earnings per share of \$1.50
- The average P/E of comparable furniture stocks is 20
- What is your estimate of the value of Herman Miller's stock using the comparables method?

Example, Valuation by Comparables

- Furniture manufacturer Herman Miller has earnings per share of \$1.50
- The average P/E of comparable furniture stocks is 20
- What is your estimate of the value of Herman Miller's stock using the comparables method?
- Solution, multiply Herman Miller's EPS by the P/E of comparable firms,
$$P = (\text{Average P/E of comparables}) \times \text{EPS} = 20 \times 1.50 = \$30$$

Example, Valuation by Comparables

- Citigroup's book value of equity is \$20 billion. It has 1 billion shares outstanding. The average market-to-book ratio of similar banks is 1.5
- Using the comparables method, what is the estimate of Citigroup's stock price?

Example, Valuation by Comparables

- Citigroup's book value of equity is \$20 billion. It has 1 billion shares outstanding. The average market-to-book ratio of similar banks is 1.5
- Using the comparables method, what is the estimate of Citigroup's stock price?
- Solution, the comparables imply an estimate of Citigroup's market capitalization,
 $M_{\text{Citigroup}} = (\text{Average M/B of comparables}) \times B_{\text{Citigroup}} = 1.5 \times 20 = \30 billion

Example, Valuation by Comparables

- Citigroup's book value of equity is \$20 billion. It has 1 billion shares outstanding. The average market-to-book ratio of similar banks is 1.5
- Using the comparables method, what is the estimate of Citigroup's stock price?
- Solution, the comparables imply an estimate of Citigroup's market capitalization,
 $M_{\text{Citigroup}} = (\text{Average M/B of comparables}) \times B_{\text{Citigroup}} = 1.5 \times 20 = \30 billion

- The question asks for the share price, not the market capitalization,

$$P = \frac{\text{Market capitalization}}{\text{Number of shares outstanding}} = \frac{\$30 \text{ billion}}{1 \text{ billion}} = \$30$$

Limitations of the Comparables Method

- The estimate is sensitive to the choice of multiple
 - E.g., P/E, market-to-book, or something else

Limitations of the Comparables Method

- The estimate is sensitive to the choice of multiple
 - E.g., P/E, market-to-book, or something else
- Good comparable companies are hard to find

Recap of Part 4

- Comparables, scale a peer multiple (P/E, market-to-book) to the firm being valued

Recap of Part 4

- Comparables, scale a peer multiple (P/E, market-to-book) to the firm being valued
- Average the comparables' multiple, then apply it to the firm's own denominator

Recap of Part 4

- Comparables, scale a peer multiple (P/E, market-to-book) to the firm being valued
- Average the comparables' multiple, then apply it to the firm's own denominator
- The estimate is sensitive to the choice of multiple, and good comparables are hard to find

Key Takeaways

- Bonds and stocks, the present value of their cash flows (coupons and face value, dividends)

Key Takeaways

- Bonds and stocks, the present value of their cash flows (coupons and face value, dividends)
- Bond prices and yields move in opposite directions

Key Takeaways

- Bonds and stocks, the present value of their cash flows (coupons and face value, dividends)
- Bond prices and yields move in opposite directions
- Prices fall and yields rise with market interest rates and with default risk, lower-rated bonds carry higher yields

Key Takeaways

- Bonds and stocks, the present value of their cash flows (coupons and face value, dividends)
- Bond prices and yields move in opposite directions
- Prices fall and yields rise with market interest rates and with default risk, lower-rated bonds carry higher yields
- Stocks, the dividend discount model or comparables

Thank you!

Please let me know if you have any questions.